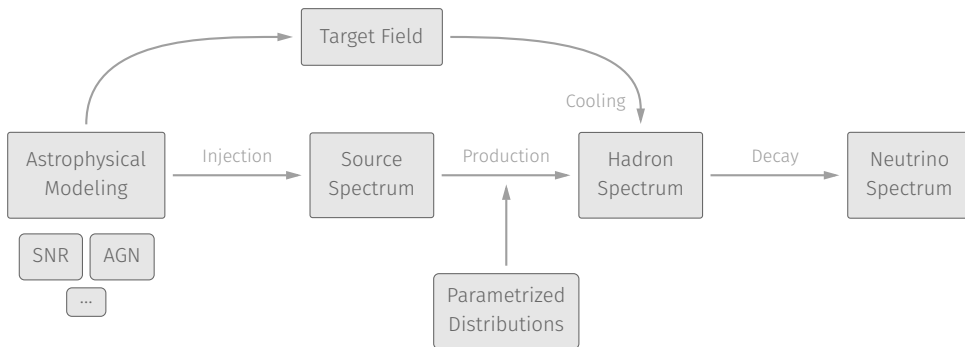

Neutrinos via Charm Decays in Astrophysical Sources

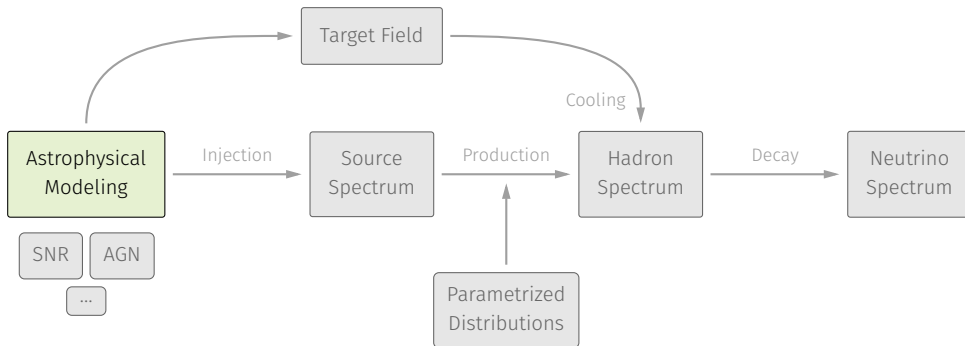
TALK IN PARTIAL FULFILLMENT OF THE REQUIREMENTS FOR THE DEGREE OF BACHELOR OF SCIENCE

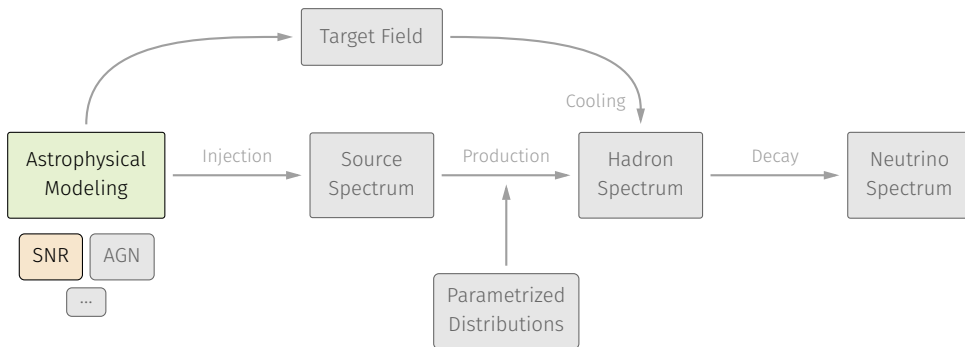
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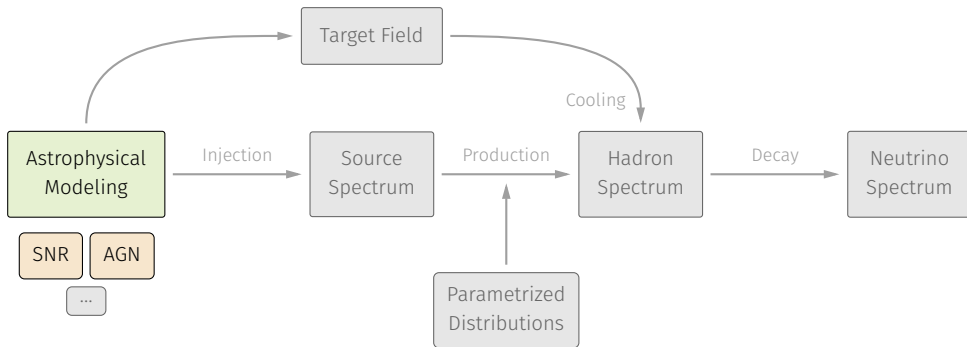
September 3, 2024

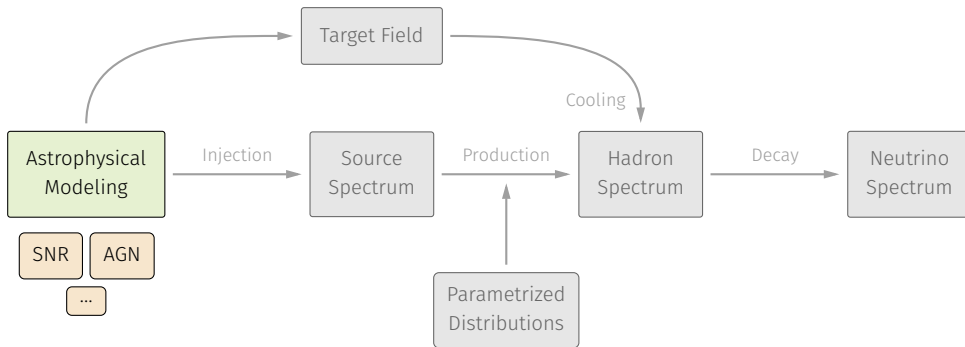
Technical University of Dortmund & Ruhr University Bochum

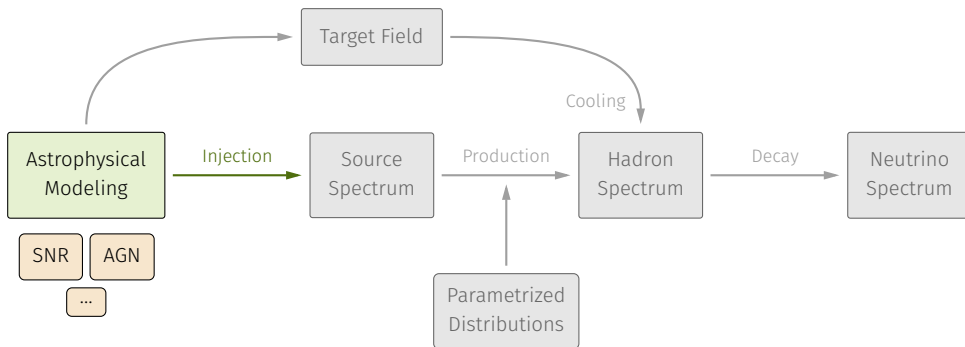


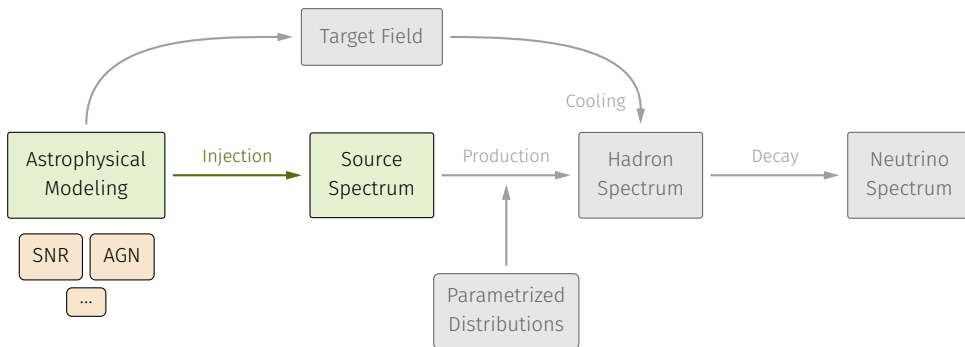


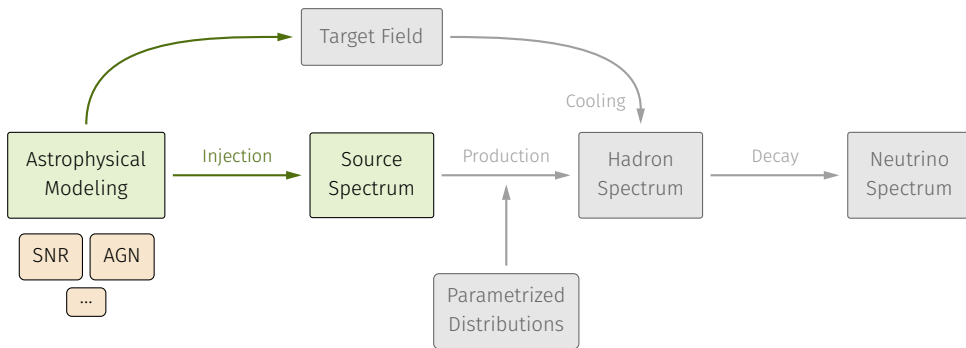


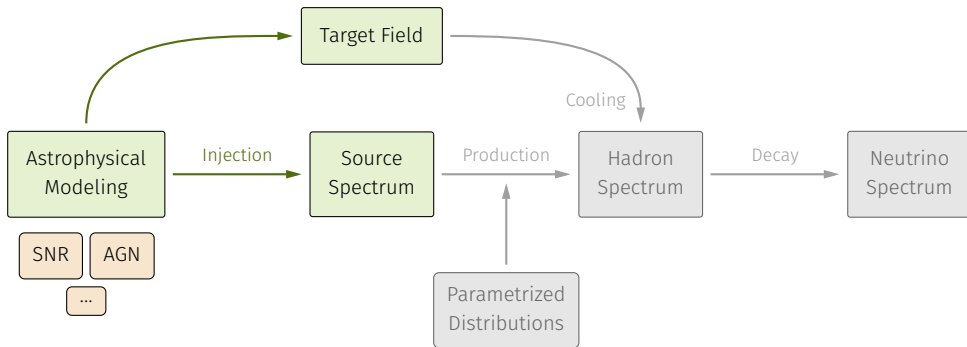


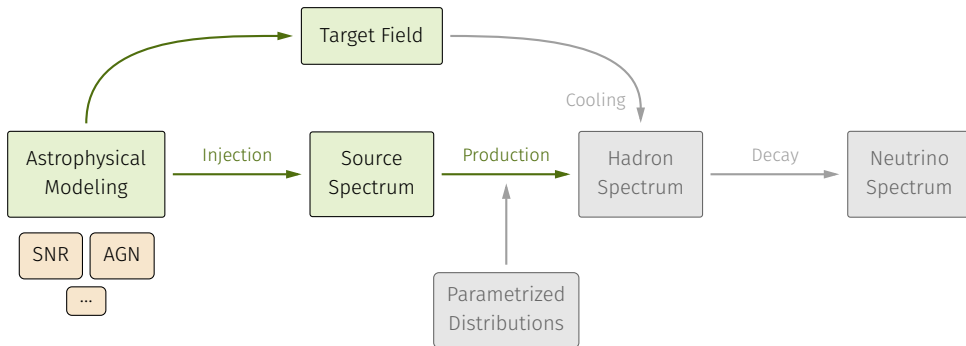


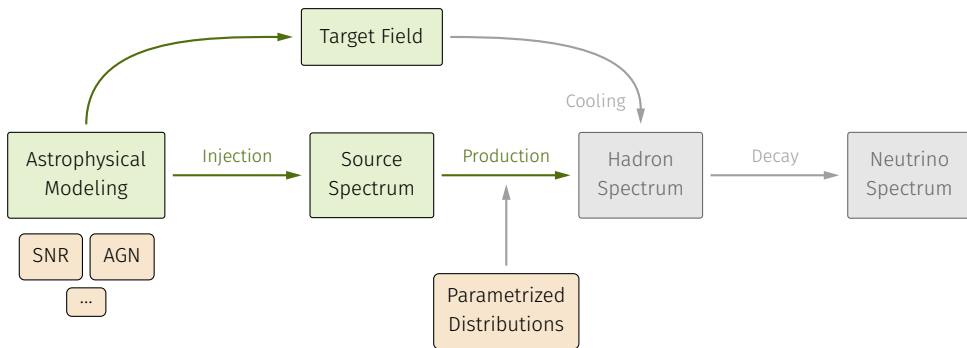


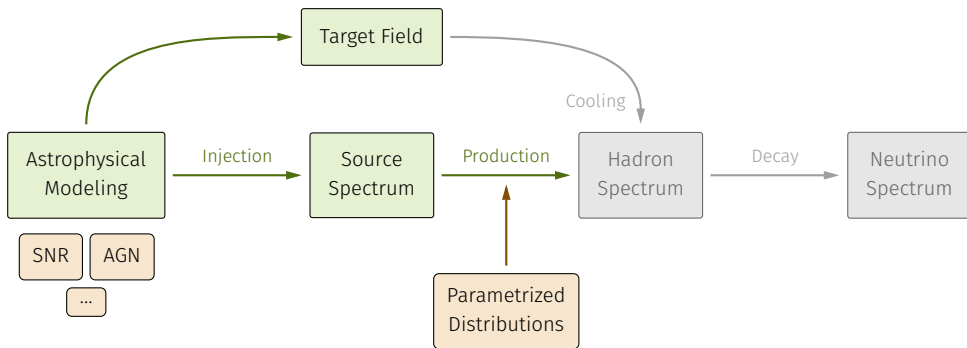


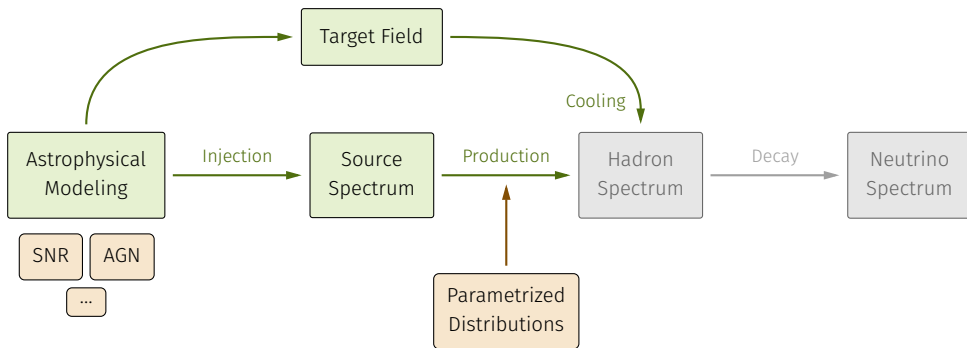


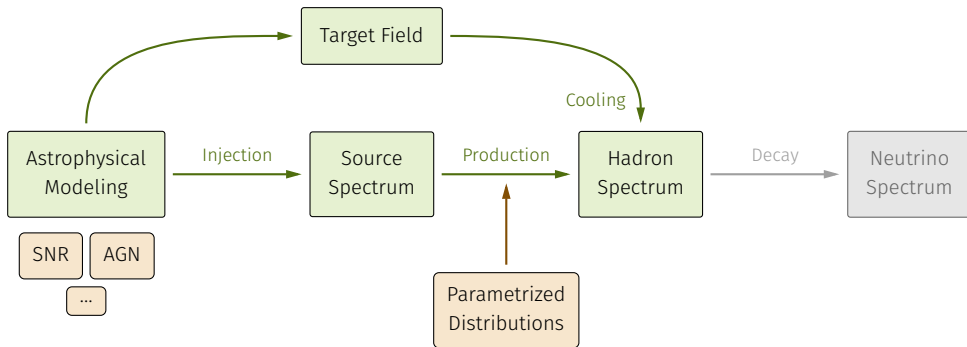


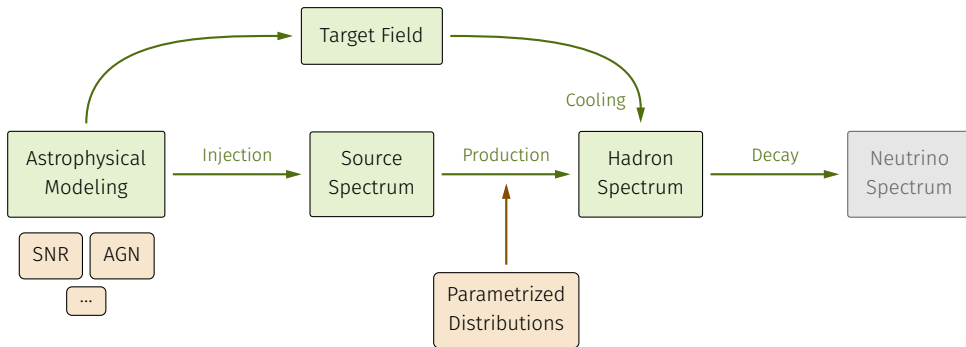


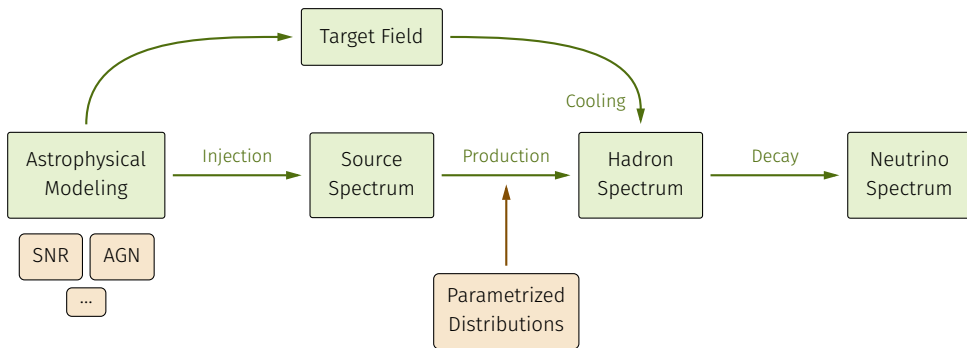


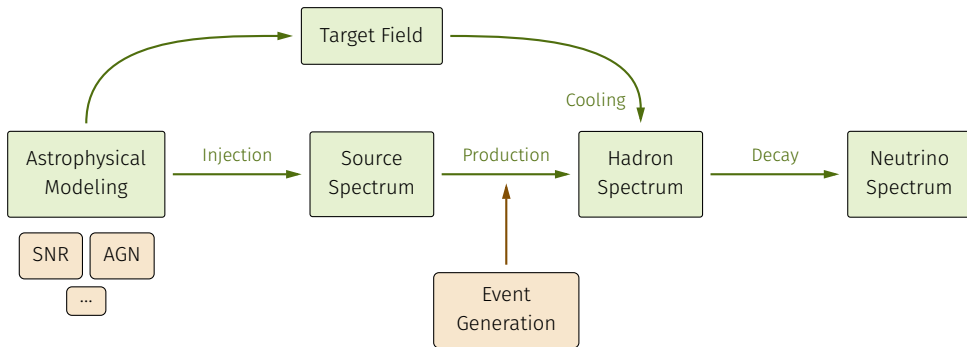




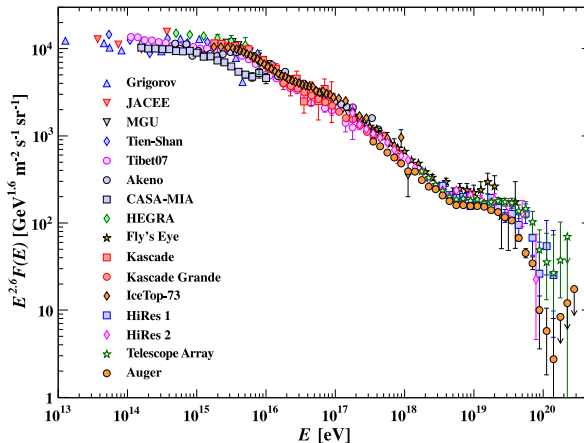






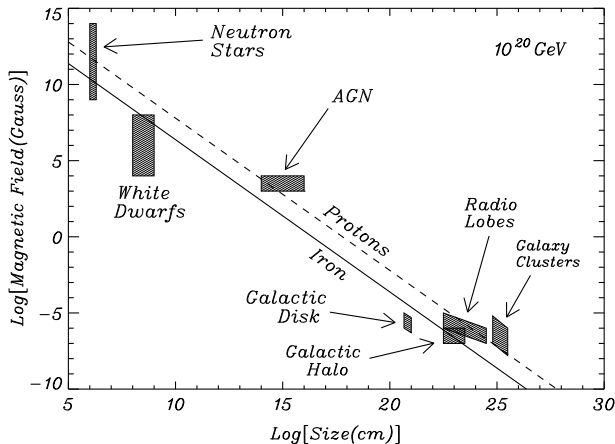


Cosmic Ray Spectrum & Cutoff



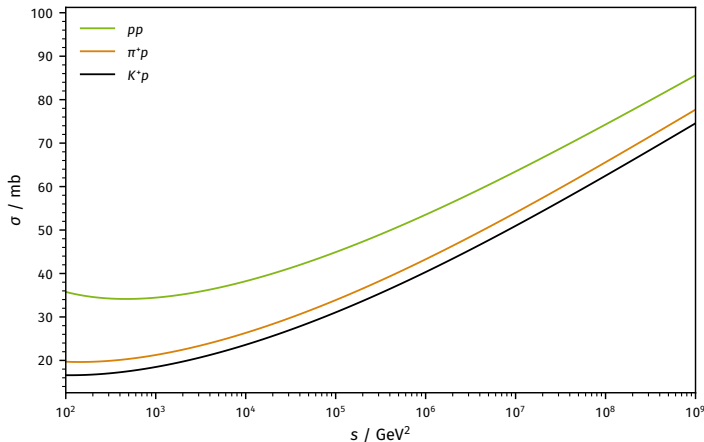
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Chin. Phys. C (2014)

Possible High Energy Cosmic Ray Origins



A. V. Olinto
AIP Conf. Proc. (2000)

Hadron Inelastic Scattering Cross Section

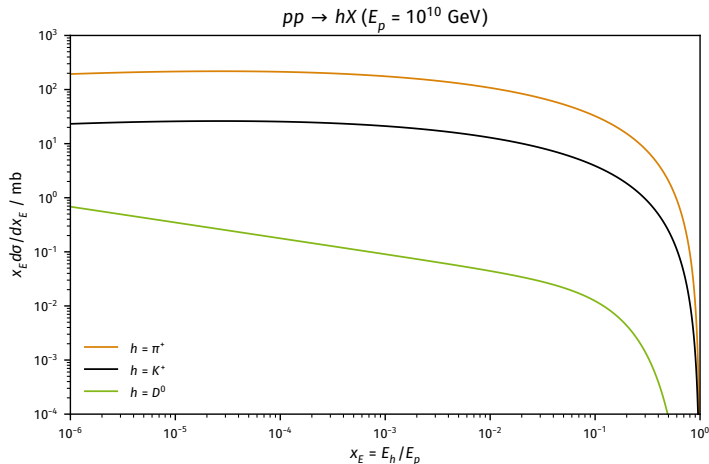


Combined Parametrizations

V. I. Belousov et. al.
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D. A. Fagundes, M. J. Menon
Nucl. Phys. A (2012)

Weighted Energy Differential Cross Section

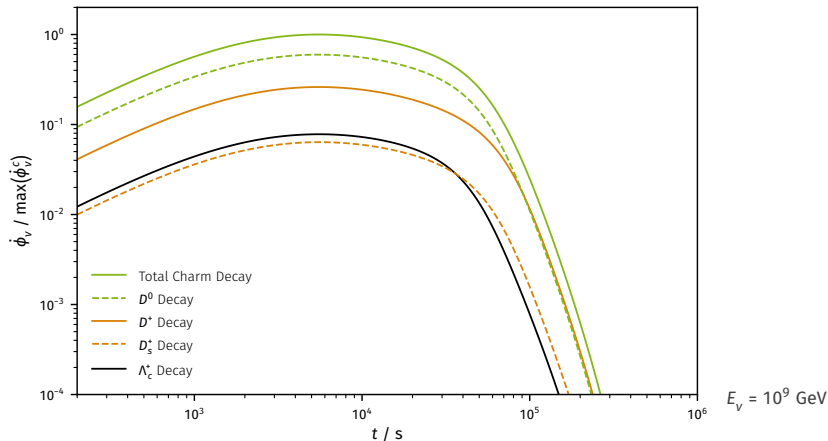


Derived Parametrizations

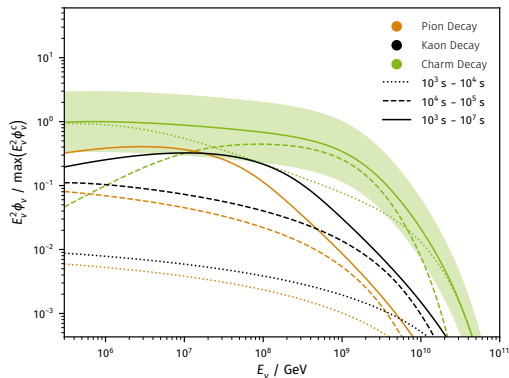
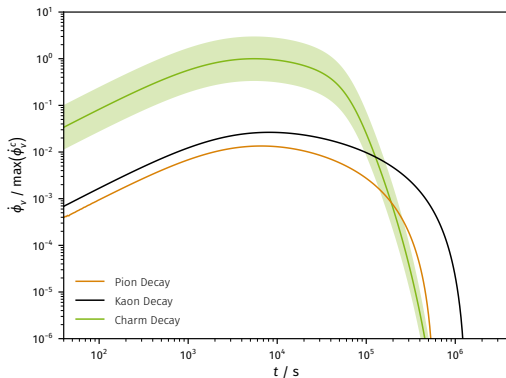
S. R. Kelner et al.
Phys. Rev. D (2006)

V. P. Gonçalves et. al.
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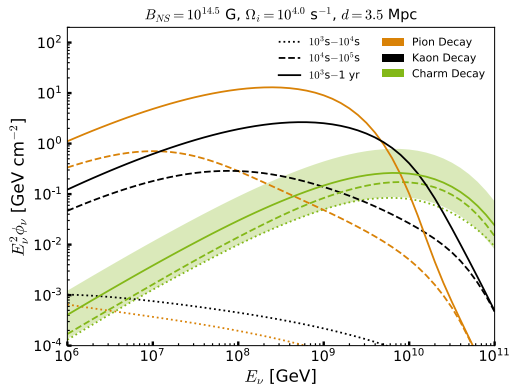
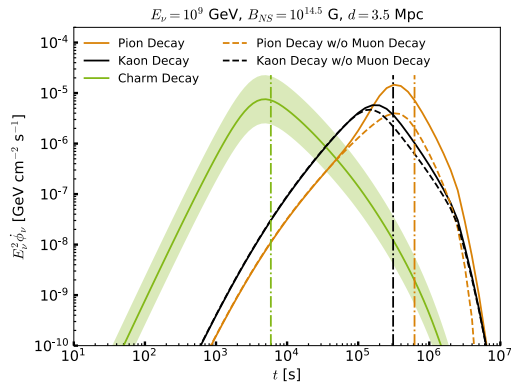
Normalized Magnetar Charm Contributions



Normalized Magnetar Neutrino Flux & Fluence

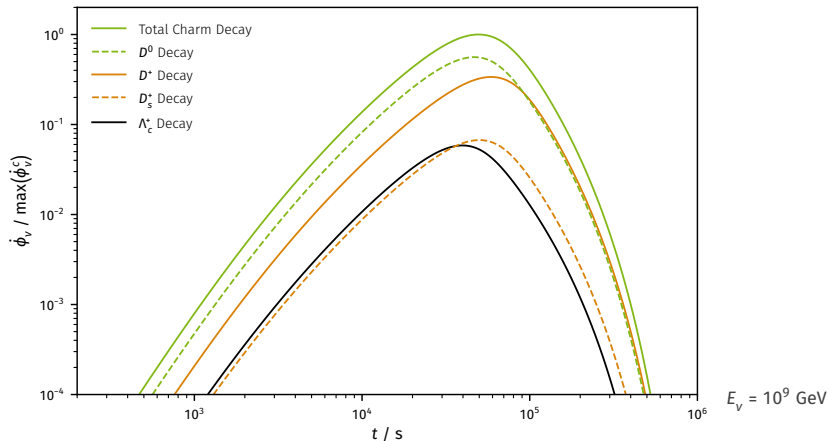


Comparison: Published Estimated Magnetar Neutrinos

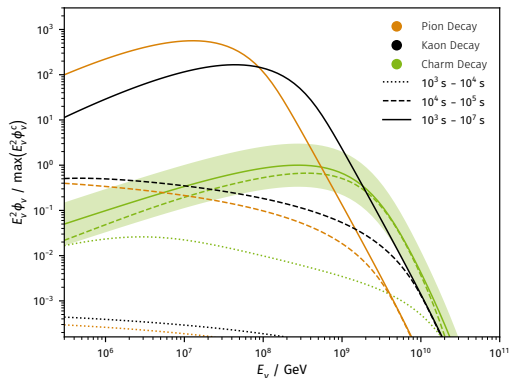
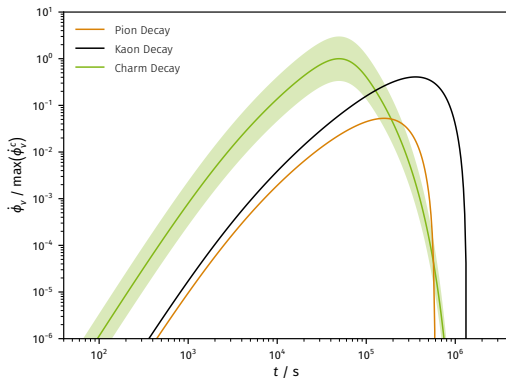


J. A. Carpio et. al. *Phys. Rev. D* (2020)

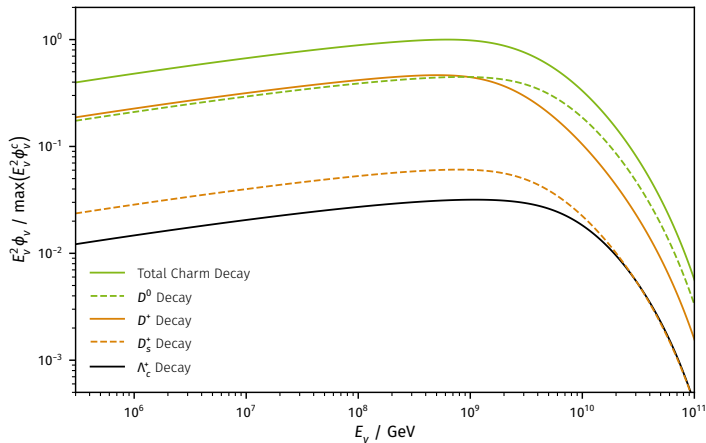
Modified Magnetar Charm Contributions



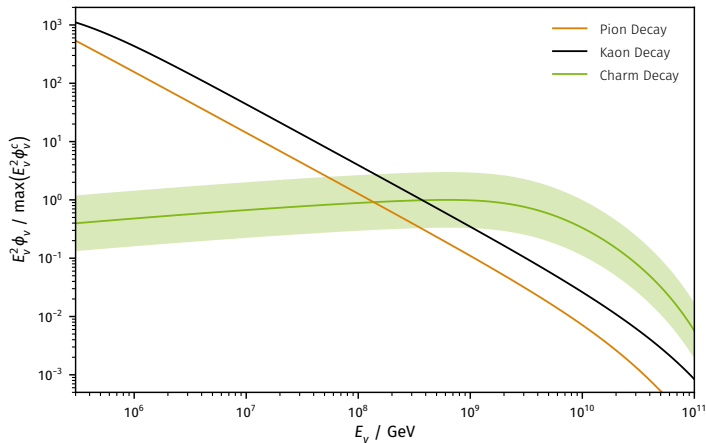
Modified Magnetar Neutrino Flux & Fluence



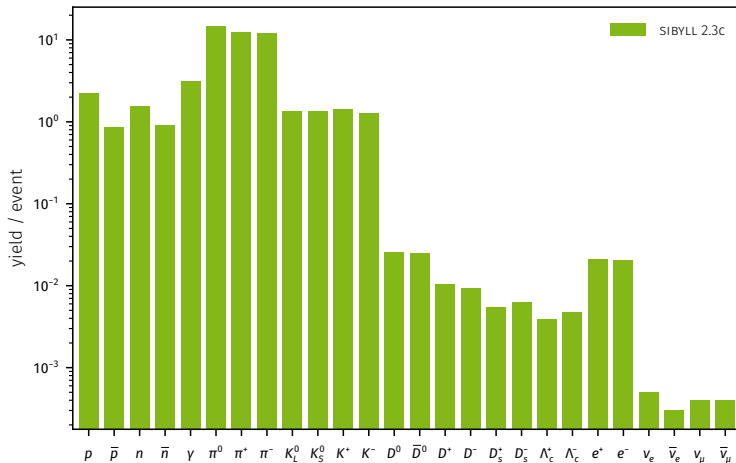
Normalized AGN Charm Contributions



Normalized AGN Neutrino Fluence

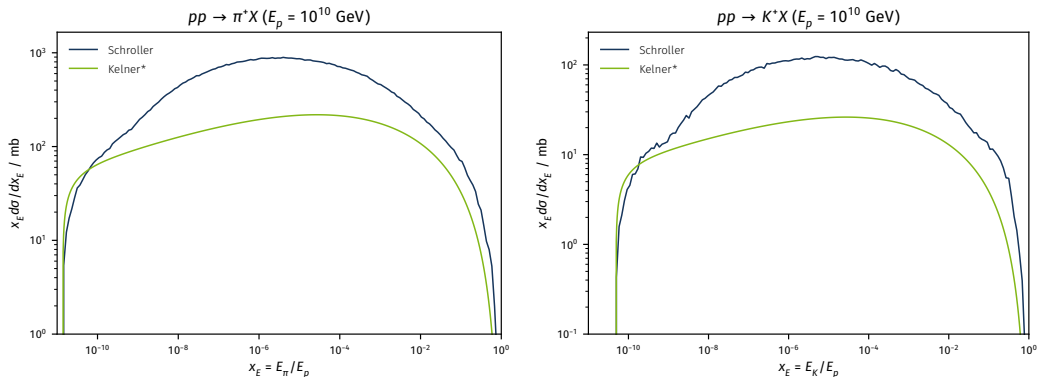


Monte Carlo Event Generation



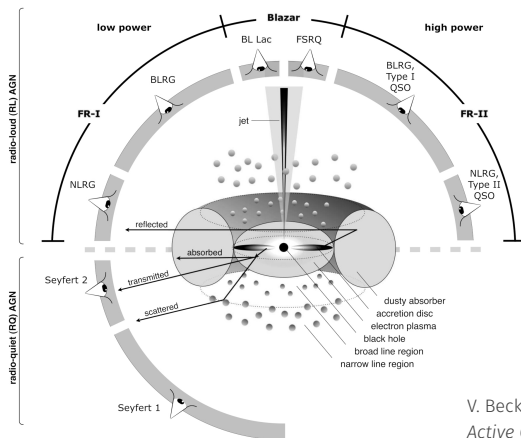
Data Courtesy
M. Schroller

Comparison: Differential Cross Section



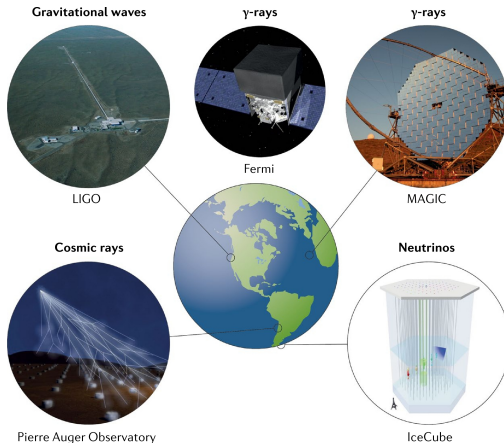
Derived Parametrization S. R. Kelner et. al. *Phys. Rev. D* (2006)
Event Data (SIBYLL 2.3c) Courtesy M. Schrollier

Unified AGN Classification Model



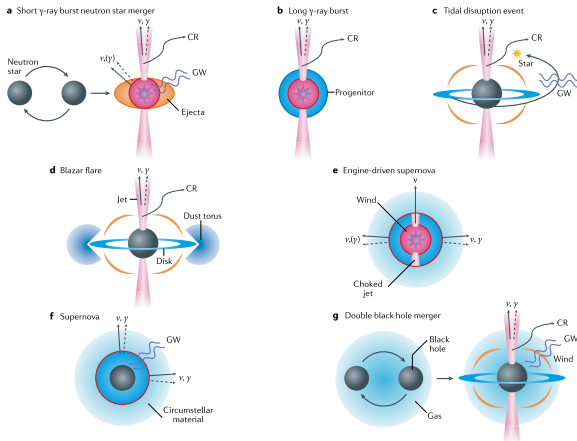
V. Beckmann, C. Shrader
Active Galactic Nuclei (2012)

Detectors



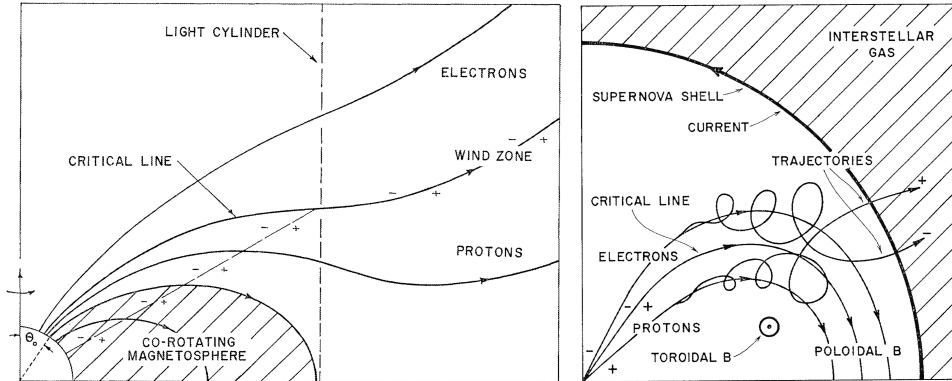
P. Mészáros et. al.
Nat. Rev. Phys. (2019)

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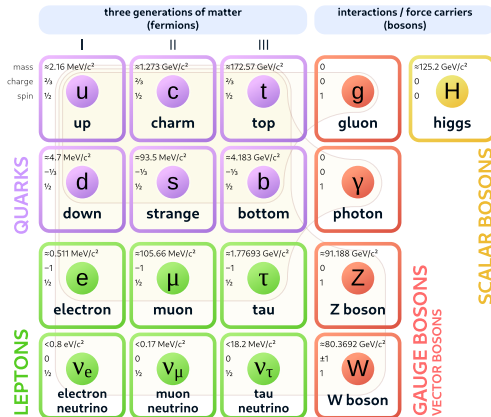
Pulsar Particle Acceleration



P. Goldreich, W. H. Julian *ApJ* (1969)

Elementary Particles

Standard Model of Elementary Particles



Standard Model
Wikimedia Commons (2019)